Chapter 9 Answers

Problem 1

Suppose that $\mathcal{C}$ contains a bluebird B and a mocker M. Thus, for all x, y and z, $Bxyz = x(yz)$ and $Mx = xx$. Now consider any elements x and y. Take C to be the combinator $(Bx)y$. Thus $(Bxy)z = ((Bx)y)z = Bxyz = x(yz)$ for every z, meaning C combines x with y, and thus the composition condition holds. By Theorem 1 of the last chapter, a fixed point of every element x is CC, where C combines x with M. Thus a fixed point of every element x in $\mathcal{C}$ is $(BxM)(BxM)$, which is the same as $M(BxM)$.

Problem 2

Suppose that $\mathcal{C}$ has some bluebird B and mocker M. Then, by Problem 2 of Chapter 8, there is an egocentric element, namely y for y the fixed point of M. In particular, this is the element $M(BMM)$. In particular: $(M(BMM))(M(BMM)) = M(M(BMM)) = (BMM)(BMM) = M(BMM)$.

Problem 3

Now suppose that $\mathcal{C}$ contains some B, M and a kestrel K. Thus by Problem 1, K has a fixed point, and hence, by Problem 20(b) of Chapter 8, the fixed point of K is narcissistic. In particular, $M(BKM)$ is narcissistic.

Problem 4

(1):

(a): Take D to be BB. Thus $Dxyzw = BBxyzw = (BBxy)zw = (B(xy))zw = B(xy)zw = (xy)(zw) = xy(zw)$.

(b): Take B_1 to be DB. Thus $B_1xyzw = DBxyzw = (DBxyz)w = (Bx(yz))w = Bx(yz)w = x((yz)w) = x(yzw)$.

(c): Take E to be BB_1. Thus $Exyzwv = BB_1xyzwv = (BB_1xy)zwv = B_1(xy)zwv = (xy)(zwv) = xy(zwv)$.

(d): Take B_2 to be EB. Thus $B_2xyzwv = EBxyzwv = (EBxyzw)v = (Bx(yzw))v = Bx(yzw)v = x((yzw)v) = x(yzwv)$.

(e): Take D_1 to be B_1B. Thus $D_1xyzwv = B_1Bxyzwv = (B_1Bxyz)wv = B(xyz)wv = (xyz)(wv) = xyz(wv)$. Alternatively, we could take D_1 to be BD. Actually, $BD = B_1B$, as $BD = B(BB) = BBBB = DBB = B_1B$. Thus $D_1xyzw = BDxyzwv = (BDxy)zwv = D(xy)zwv = (xy)z(wv) = xyz(wv)$.

(f): Take B^+ to be D_1B. Thus $B^+xyzw = D_1Bxyzw = Bxy(zw) = x(y(zw))$.

(g): Take D_2 to be DD. Thus $D_2xyzwv = DDxyzwv = (DDxyz)wv = Dx(yz)wv = x(yz)(wv)$.

(h): Take $\hat{E}$ to be EE. Thus

$$\hat{E}xy_1y_2y_3z_1z_2z_3 = EExy_1y_2y_3z_1z_2z_3 = (EExy_1y_2y_3)z_1z_2z_3 = Ex(y_1y_2y_3)z_1z_2z_3 = x(y_1y_2y_3)(z_1z_2z_3)$$

(2): The base case, when $B = 0$, is B itself. Now suppose that the hypothesis holds for some n, that is,

$$B_nxy_1...y_{n+2} = x(y_1...y_{n+2})$$

Let B_{n+1} be the combinator BB_nB. Thus

$$BB_nBxy_1...y_{n+3} = (BB_nBx)y_1...y_{n+3} = B_n(Bx)y_1...y_{n+3} = (B_n(Bx)y_1...y_{n+2})y_{n+3} = (Bx)(y_1...y_{n+2})y_{n+3}$$

$$= Bx(y_1...y_{n+2})y_{n+3} = x((y_1...y_{n+2})y_{n+3}) = x(y_1...y_{n+2}y_{n+3})$$

thus showing that B_{n+1} is the desired combinator, and therefore the proof is complete. Alternatively, as Smullyan gives in his answer, it can be seen that taking B_{n+1} to be BBB_n also works.

Problem 5

Suppose that we have a truth T and a lark L. Thus $Txy = yx$ and $Lxy = x(yy)$ for all x and y. Now, by Problem 23 of Chapter 7, T has a fixed point. Let A be some fixed point of T. Thus $Ax = TAx = xA$, meaning A commutes with every element x.

Problem 6

Suppose we have a bluebird B and a trush T. Let R be the combinator DT (that is, BBT), meaning $Rxyz = DTxyz = Tx(yz) = (yz)x = yzx$.

Problem 7

Take C to be RRR. Thus $Cxyz = RRRxyz = RxRyz = Ryxz = xzy$.

Problem 8

Here is a reduction of C from Problem 7 to B and T, using only 8 letters: $C = RRR = DT(DT)(DT) = BBT(BBT)(BBT) = B(T(BBT))(BBT)$.

Problem 9

Suppose we have a cardinal C and an indentity combinator I. Take T to be CI. Thus $Txy = CIxy = Iyx = yx$, meaning T is a thrush.

Problem 10

Suppose we have some cardinal C. Take R to be CC. Thus $Rxyz = CCxyz = Cyxz = yzx$, meaning R is a robin.

Problem 11

(a): Take F to be BCR, meaning $Fxyz = BCRxyz = C(Rx)yz = Rxzy = zyx$, meaning F is a finch.

(b): Take F to be $ETTET$, meaning $Fxyz = ETTETxyz = TT(ETx)yz = (ETx)Tyz = ETxTyz = Tx(Tyz) = (Tyz)x = Tyzx = zyx$.

Problem 12

Take V to be CF, meaning $Vxyz = CFxyz = Fyxz = zxy$, meaning V is a vireo.

Problem 13

Take F to be CV, meaning $Fxyz = CVxyz = Vyxz = zyx$.

Problem 14

Suppose that we have a robin R and a kestrel K. Take I to be RKK. Thus $Ix = RKKx = KxK = x$, meaning I is an identity combinator. In fact, we can take I to be RAK for any element A, as $RAKx = KxA = x$.

Problem 15

(a): Take C^* to be BC, meaning $C^*wxyz = BCwxyz = C(wx)yz = (wx)zy = wxzy$.

(b): Take R^* to be BCC^* (that is, C^*C^*), meaning $R^*wxyz = BCC^*wxyz = C(C^*w)xyz = C^*wyxz = wyzx$.

(c): Take F^* to be BC^*R^*, meaning $F^*wxyz = BC^*R^*wxyz = C^*(R^*w)xyz = R^*wxzy = wzyx$.

(d): Take V^* to be C^*F^*, meaning $V^*wxyz = C^*F^*wxyz = F^*wyxz = wzxy$.

Problem 16

For all parts of this problem, take the new combinator A^{**} to be the combinator BA^*. For example, in the case of C^{**}, we have $C^{**}vwxyz = BC^*vwxyz = C^*(vw)xyz = vwxzy$. More generally, if $Axyz = abc$, where abc is a permutation of xyz, then $BA^*wvxyz = A^*(wv)xyz = wvabc$.

Note that this problem can be generalized in the following way: If $Axyz = abc$ for some permutation abc of xyz, let A_n for $n > 1$ be defined in the following way: $A_nx_1...x_nxyz = x_1...x_nxyz$. If $n = 2$, let $A_n = BA^*$, which means $A_nvwxyz = vwxyz$ by Problem 16. If $n > 2$, let $A_n = BA_{n-1}$, and because $BA_{n-1}x_1...x_nxyz = A_{n-1}(x_1x_2)...x_nxyz = (x_1x_2)x_3...x_nabc = x_1x_2...x_nabc$, A_n satisfies the desired condition.

nth Identity Combinator

Assuming a bluebird B and an identity combinator I, we can define, for every n, a combinator I_n such that $I_nx_1...x_{n+1} = x_1...x_{n+1}$. The base case when $n = 0$ is simple: we take I_0 to be I. Now suppose $n > 0$. Then take I_n to be $B_{n-1}I$. Thus

$$I_nx_1...x_{n+1} = B_{n-1}Ix_1...x_{n+1} = I(x_1...x_{n+1}) = x_1...x_{n+1}$$

By Problem 14, the nth Identity Combinator is also present if some robin R (or equivalently, some cardinal C) and some kestrel K are present (and of course, also a bluebird B).

Problem 17

Let V be C^*T. Thus $Vxyz = C^*Txyz = Txzy = zxy$, meaning V is a vireo. Expanded to B and T, this is: $V = C^*T = (BC)T = (B(RRR))T = (B(B(T(BBT))(BBT)))T$ (the last step uses Problem 8).

Problem 18

Just take A to be T: $Axyz = Txyz = (Txy)z = (yx)z = yxz$.

Problem 19

Consider the A from Problem 18. Let A^* be the combinator BR^*C^*. Thus $A^*wxyz = BR^*C^*wxyz = R^*(C^*w)xyz = C^*wyzx = wyxz$. Then take Q to be A^*Bxyz, meaning $Qxyz = A^*Bxyz = Byxz = y(xz)$. Note that by Problem 16, $A^{**} = BA^*$.

A far simpler solution is taking Q to be CB, as $CBxyz = Byxz = y(xz)$.

Problem 20

Take Q_1 to be C^*B, meaning $Q_1xyz = C^*Bxyz = Bxzy = x(zy)$. Take Q_2 to be R^*B, meaning $Q_2xyz = R^*Bxyz = Byzx = y(zx)$.

Note that, in general, if Q_i is defined to be $Q_ixyz = a(bc)$ for some permutation abc of xyz, we can take Q_i to be A^*B where $Axyz = abc$, as $A^*Bxyz = Babc = a(bc)$. Given a combinator Q_i such that $Q_ixyz = a(bc)$ for some permutation abc of xyz, define Q_i^* to be the combinator satisfying the condition $Q_i^*wxyz = wa(bc)$. Take Q_1^* to be BQ_1, as $BQ_1wxyz = Q_1(wx)yz = (wx)(zy) = wx(zy)$. Take Q_2^* to be $A^{**}Q_1^*$ with the A that was defined in Problem 18, as $A^{**}Q_1^*wxyz = Q_1^*wyxz = wy(zx)$. Let Q_0 be an abbreviation for Q, the queer bird. Take Q_0^* to be $R^{**}Q_1^*$, as $R^{**}Q_1^*wxyz = Q_1^*wyzx = wy(xz)$. In a similar fashion, take Q_3^* to be $F^{**}Q_1^*$ (as $F^{**}Q_1^*wxyz = Q_1^*wzyx = wz(xy)$), and Q_4^* to be $V^{**}Q_1^*$ (as $V^{**}Q_1^*wxyz = Q_1^*wzxy = wz(yx)$).

We can generalize the concept of Q_i^* in a near identical fashion to that expressed in Problem 16. First, let Q_i^* be Q_i^1. Now, we want a combinator Q_i^n, for all $n > 0$, such that $Q_i^nx_1...x_nxyz = x_1...x_na(bc)$ where abc is a permutation of xyz and $Q_ixyz = a(bc)$. The base cases were proven in the last paragraph (as Q_i^1 is Q_i^*). Now if $n > 1$, we, just as in Problem 16, take Q_i^n to be BQ_i^{n-1}, as $BQ_i^{n-1}x_1...x_nxyz = Q_i^{n-1}(x_1x_2)...x_nxyz = (x_1x_2)...x_na(bc) = x_1x_2...x_na(bc)$.

Problem 21

Suppose that a cardinal C is present. On the one hand, if a quixotic bird Q_1 is present, take Q_2 to be CQ_1, as $CQ_1xyz = Q_1yxz = y(zx)$. Alternatively, take Q_2 to be CRQ_1 (recalling that R is just CC), as $CRQ_1xyz = RxQ_1yz = Q_1yxz = y(zx)$, meaning Q_2 is quizzied. On the other hand, if a quizzied bird Q_2 is present, take Q_1 to be CQ_2, as $CQ_2xyz = Q_2yxz = x(zy)$, meaning Q_1 is quixotic.

Problem 22

By the note in Problem 20, we can take Q_3 to be V^*B, and Q_4 to be F^*B. Alternatively, we could take Q_3 to be BT, as $BTxyz = T(xy)z = z(xy)$, or QQC, as $QQCxyz = C(Qx)yz = Qxzy = z(xy)$, or $BC(CB)$, as $BC(CB)xyz = C(CBx)yz = CBxzy = Bzxy = z(xy)$; by Problem 23, we could also take Q_4 to be CQ_3. Take Q_5 to be V^*B_1, as $V^*B_1xyzw = B_1zxyw = z(xyw)$. Alternatively, we could take Q_5 to be BQ, as $BQxyzw = Q(xy)zw = z((xy)w) = z(xyw)$. Take Q_6 to be $C^{**}Q_5$, as $C^{**}Q_5xyzw = Q_5xywz = w(xyz)$.

Problem 23

Suppose that a cardinal C is present. On the one hand, if a quirky bird Q_3 is present, take Q_4 to be CQ_3, as $CQ_3xyz = Q_3yxz = z(yx)$. On the other hand, if a quacky bird Q_4 is present, take Q_3 to be CQ_4, as $CQ_4xyz = Q_4yxz = z(xy)$.

Problem 24

Take Q_4 to be Q_1T, as $Q_1Txyz = T(yx)z = z(yx)$.

Problem 25

Take B to be the combinator $QT(QQ)$, as $QT(QQ)xyz = QQ(Tx)yz = Tx(Qy)z = Qyxz = x(yz)$, meaning B is a bluebird.

Problem 26

Take C to be $QQ(QT)$, as $QQ(QT)xyz = QT(Qx)yz = Qx(Ty)z = Ty(xz) = xzy$, meaning C is a cardinal.

Problem 27

Take G to be $V^{**}D$, as $V^{**}Dxyzw = Dxwyz = xw(yz)$.

Alternatively, we could take, as Smullyan does, G to be BBC, as $BBCxyzw = B(Cx)yzw = Cx(yz)w = xw(yz)$.

Problem 28

Take M_2 to be $BMxy$, as $BMxy = M(xy) = (xy)(xy) = xy(xy)$.

In general, if we want a $M_nx_1...x_n = x_1...x_n(x_1...x_n)$ for $n \geq 2$, we can let $M_n = B_{n-2}M$ (letting $B_0 = B$), as $B_{n-2}Mx_1...x_n = M(x_1...x_n) = x_1...x_n(x_1...x_n)$.

Problem 29

We could take W' to be M_2R, as $M_2Rxy = Rx(Rx)y = Rxyx = yxx$. This is an expression of five letters when reduced to B, T and M, as $M_2R = (BM)(DT) = BM(BBT)$. Alternatively, we could take W'

to be $B(M_2B)T$, as $B(M_2B)Txy = M_2B(Tx)y = B(Tx)(B(Tx))y = Tx(B(Tx)y) = B(Tx)yx = Tx(yx) = yxx$. This is an expression of five letters when reduced to B, T and M, as $B(M_2B)T = B(BMB)T$.

Problem 30

Take W to be CW', as $CW'xy = W'yx = xyy$. This, when reduced to B, T and M, is

$$CW' = B(T(BBT))(BBT)(BM(BBT))$$

by Problem 8, which is an expression of 13 letters. We could also take W to be

$$B(T(BBT))(BBT)(B(BMB)T)$$

as $B(BMB)T$ is an alternative form of W'. The first expression can be simplified in the following way:

$$B(T(BBT))(BBT)(BM(BBT)) = T(BBT)(BBT(BM(BBT))) = BBT(BM(BBT))(BBT)$$

$$= B(T(BM(BBT)))(BBT)$$

which is an expression of ten letters. The second expression can be simplified in the following way:

$$B(T(BBT))(BBT)(B(BMB)T) = T(BBT)(BBT(B(BMB)T)) = BBT(B(BMB)T)(BBT)$$

$$= B(T(B(BMB)T))(BBT)$$

Note that a couple of generalizations can be made about the warbler. First, we want a combinator W^n such that $W^nxy = xy_1...y_{n+1}$ where $y_i = y$ for every i. W^1 is just W. Let $W^2 = W_2BW$, as $W_2BWxy = BWxyy = W(xy)y = (xy)yy = xyyy$. We can generalize this concept with induction. Suppose the hypothesis holds for some n, that is, $W^nxy = xy_1...y_{n+1}$. Then let $W^{n+1} = W_2BW^n$, as $W_2BW^nxy = BW^nxyy = W^n(xy)y = xyy_1...y_{n+1} = xy_1...y_{n+2}$ where every $y_i = y$. Now, let $W_0 = W$. We want a combinator W_n^m such that $W_n^m z_1...z_{n+1}w = z_1..z_{n+1}w_1...w_{m+1}$ for all $n \geq 1$. For the case of W_0^n, we simply define W_0^n to be W^n for every n. We already have a combinator W^n such that $W^nxy = xy_1...y_{n+1}$. Take W_n^m to be $B_{n-1}W^m$, as $B_{n-1}W^mz_1...z_{n+1}w = W^m(z_1...z_{n+1})w = z_1...z_{n+1}w_1...w_{m+1}$.

Problem 31

Take M to be WT, as $WTx = Txx = xx$. Second, take M to be WI, as $WIxx = Ixx = xx$. Third, take I to be WK, as $WKx = Kxx = x$.

Problem 32

Take W_1 to be BW, as $BWxyz = W(xy)z = (xy)zz = xyzz$. Take W_2 to be $B_1Wxyzw = W(xyz)w = xyzww$. Take W_3 to be $B_2Wxyzwv = W(xyzw)v = xyzwvv$.

In general, if we want $W_nx_1...x_{n+2} = x_1...x_{n+1}x_{n+2}x_{n+2}$, we can take W_n to be $B_{n-1}W$ (letting $B = B_0$), as $B_{n-1}Wx_1...x_{n+2} = W(x_1...x_{n+1})x_{n+2} = x_1...x_{n+1}x_{n+2}x_{n+2}$. Smullyan provides an alternative way of defining W_n. In paritcular, let $W_0 = W$. Then, if $W_nx_1...x_{n+2} = x_1...x_{n+1}x_{n+2}x_{n+2}$, take W_{n+1} to be BW_n, as $BW_nx_1...x_{n+3} = W_n(x_1x_2)...x_{n+3} = x_1x_2...x_{n+3}x_{n+3}$.

Problem 33

Take H to be W_1C^*, as $W_1C^*xyz = C^*xyyz = xyzy$.

Problem 34

Take W to be $C(HR)$, as $C(HR)xy = HRyx = Ryxy = xyy$. We could also take HR to be W', as $HRxy = Rxyx = yxx$, meaning W could be simplified to CW'. Note that because $C = RRR$, we could expand W to $RRRW'$.

Problem 35

(a): Assume the combinators B, R and M. Take L to be RMB, as $RMBxy = BxMy = x(My) = x(yy)$. Now assume the combinators B, C and M. Take L to be CBM, as $CBMxy = BxMy = x(My) = x(yy)$. This can be reduced to B, M and T in the following way: $CBM = (B(T(BBT))(BBT))BM$ by Problem 8.

(b): Assume the combinators B and W. Take L to be W_1B, as $W_1Bxy = Bxyy = x(yy)$.

(c): Let L be QM, as $QMxy = x(My) = x(yy)$. When reduced to B, M and T, this is: $QM = CBM = (B(T(BBT))(BBT))BM$ by Problem 8.

Problem 36

Take S to be W_2G, as $W_2Gxyz = Gxyzz = xz(yz)$. This can be reduced to: $W_2G = B(BW)(BBC)$.

Problem 37

Take H to be SR, as $SRxyz = Ry(xy)z = xyzy$.

Problem 38

Suppose we have S and T. Take W to be ST, as $STxy = Ty(xy) = xyy$. Take M to be STT, as $STTx = Tx(Tx) = Txx = xx$.

Problem 39

Take W' to be SRR, as $SRRxy = Rx(Rx)y = Rxyx = yxx$. Then, by Problem 30, CW' is a warbler. In terms of C and S, this is:

$$C(SRR) = C(S(CC)(CC))$$

Problem 40

(a): Take P to be DW_1Q, as $DW_1Qxyz = W_1Q(xy)z = Q(xy)zz = z(xyz)$. Alternatively, we could, as Smullyan does, take P to be W_2Q_5, as $W_2Q_5xyz = Q_5xyzz = z(xyz)$.

(b): Take P_1 to be W_1BQ, as $W_1BQxyz = BQxxyz = Q(xx)yz = y(xxz)$. Alternatively, we could take P_1 to be LQ, as $LQxyz = Q(xx)yz = y(xxz)$.

(c): Take P_2 to be W_1B_1, as $W_1B_1xyz = B_1xyyz = x(yyz)$. Alternatively, we could take P_2 to be CP_1, as $CP_1xyz = P_1yxz = x(yyz)$.

(d): Take P_3 to be DW_1Q_5C, as $DW_1Q_5Cxyz = W_1Q_5(Cx)yz = Q_5(Cx)yyz = y(Cxyz) = y(xzy)$. More simply, take P_3 to be BCP, as $BCPxyz = C(Px)yz = Pxzy = y(xzy)$.

Problem 41

Let M be PII, as $PIIx = x(IIx) = x(x) = xx$.

Problem 42

(a): Take Φ to be B_1SB, as $B_1SBxyzw = S(Bxy)zw = Bxyw(zw) = x(yw)(zw)$. Alternatively, we could, as Smullyan does, take Φ to be $B(BS)B$, as $B(BS)Bxyzw = BS(Bx)yzw = S(Bxy)zw = Bxyw(zw) = x(yw)(zw)$.

(b): The base case, where $n = 1$, is part (a). Now assume the hypothesis holds for some n, that is:

$$\Phi_n xyzw_1...w_n = x(yw_1...w_n)(zw_1...w_n)$$

Take Φ_{n+1} to be $B\Phi_n\Phi$, as

$$B\Phi_n\Phi xyzw_1...w_{n+1} = \Phi_n(\Phi x)yzw_1...w_{n+1} = \Phi x(yw_1...w_n)(zw_1...w_n)w_{n+1}$$

$$= x(yw_1...w_nw_{n+1})(zw_1...w_nw_{n+1})$$